

Robust Trust Theorem 2 infinite extension: A chronicle of attempted routes

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Abstract

This is a research log of the routes attempted to extend the existence direction of Theorem 2 (“Robustly Rationalizable Solution”) in Dworczak & Smolin’s *Robust Trust* (2026) to infinite M and Θ . It documents what was tried, what worked, what failed, and where the failures share a common structural cause. The companion document *exposition.pdf* states the resulting three-tier theorem and its sharpness; this chronicle exists so that future contributors do not redo failed work and so that the central remaining open problem is precisely posed. Section 2 records the route based on Phil Reny’s restricted-game suggestion in detail; subsequent sections record the menu-engine pivot and the post-v8 strategic search.

1 The question and the gap

The paper’s Theorem 2 has two parts. *Optimality direction*: every robustly rationalizable strategy is optimal. *Existence direction*: when $M = \text{supp}(\tau)$ and Θ are finite, a robustly rationalizable strategy exists. The optimality direction is finiteness-free; the existence direction uses Sion’s minimax on products of finite-dimensional simplices and the paper notes that extending Sion’s conditions to infinite cheap-talk-like spaces is technically difficult.

The project asked: can existence be established in infinite M and compact metric Θ , under the model’s standing Borel/compact hypotheses? The answer reached over the course of the routes below is that a clean three-tier conditional theorem can, but the unrestricted existence direction remains structurally open at a single named obstacle: a deletion-compatible Hall duality theorem (see §6).

2 The Phil Reny route

The first sustained attempt followed Phil Reny’s suggestion: split the proof into a value-securing existence step and an attainment step, restrict the misaligned adviser to τ -dominated kernels in the first step, then lift the restricted maximizer back to the unrestricted game via Lusin regularization.

Stage 1 — restricted-game maximizer. Restrict B to the convex set F of τ -dominated densities $\beta_\varphi(dm \mid s) = \varphi(m \mid s)\tau(dm)$ with φ jointly Borel and normalized. On the restricted game, two ingredients combine cleanly:

- **Balder constant-marginal continuity.** For every fixed $\varphi \in F$, the map $\sigma \mapsto U_F(\sigma, \varphi) := U(\beta_\varphi, \sigma)$ is continuous in the Balder weak topology T_λ (Balder 1988, Theorem 2.2), where $\lambda := \tau \otimes \bar{f}$ with $\bar{f} := \sum_\omega \mu_0(\omega)f(\cdot \mid \omega)$.

- **Compactness of Σ in T_λ .** Modulo the λ -a.e. equivalence (Balder quotient), Σ is compact convex Hausdorff (Balder 1988, Theorem 2.3(a)).

Mertens’s asymmetric minmax theorem (Mertens 1986, Corollary B) then gives a restricted-game maximizer $\sigma^* \in \Sigma$ with $\max_\sigma \inf_F U_F = \inf_F \max_\sigma U_F$. Both mixed-strategy sides collapse to pure by affineness and the standard barycenter argument.

Stage 2 — Lusin lift. For each $\beta \in B$ and $\varepsilon > 0$, smooth β to a τ -dominated $\varphi_\varepsilon \in F$ losing at most ε at σ^* . The smoothing is built shell-by-shell on a Lusin compact core $K^* = \bigcup_n K_n$ with full $\pi(\cdot | \omega)$ -measure for every ω , where $\hat{\sigma}^*$ is continuous on each K_n in the Balder stable topology and every relative open in K_n has positive $\pi(\cdot | \omega)$ -mass.

Hypotheses required. Three hypotheses entered:

1. **(A5-thick)** the Lusin-thickness condition just stated. The original (A5) demanded $\pi(\cdot | \omega) \sim \tau$ for every ω ; (A5-thick) is strictly weaker.
2. **(A8c-attain)** the rowwise infimum of the misaligned payoff $\ell_{\sigma^*}(m, s) := \sum_\omega s(\omega) p_\omega(m)$ is attained at a measurable selector. This converts the value-securing ε -adversary into an exact β^* .
3. **(TRE-gen-Hall)** a deterministic version of the posterior-calibration condition needed for full robust rationalizability: the joint coupling $\gamma_\alpha = \alpha(\text{id}, \text{id})_\# \tau + (1 - \alpha)(\text{id}, m^*)_\# \tau$ disintegrates with posteriors landing in the per-message Bayes cones $C(m)$ for q -a.e. m , where m^* is the deterministic worst-message map.

Result and limitation. This delivered a two-tier conditional theorem (Tier 1: optimal σ^* and adversarial β^* with $U(\sigma^*) = U^* = U(\beta^*, \sigma^*)$; Tier 2: full robust rationalizability under the third hypothesis). Each of the three added conditions was shown *tight for the route*: a perfect-revelation finite example obstructs the Lusin-smoothing mechanism behind A5-thick, an upward-spike row obstructs A8c-attain, and a ternary non-radial witness exhibits an explicit Hall violation showing TRE-gen-Hall is not derivable from the weaker hypotheses.

The route was real, but the engine was heavy. (A5-thick) was imposed on a representative of a maximizer of an abstract infinite-dimensional optimization, and (A8c-attain) inherited that representative-dependence. The witness for (TRE-gen-Hall) hinted at something more fundamental than a deterministic-vs-randomized issue, but the route’s deterministic kernel architecture made that hard to disentangle.

3 The pivot to the menu engine

The architectural change came from observing that the misaligned payoff at fixed σ depends on the agent’s strategy only through its image in the *payoff-profile space*

$$W := \{w \in \mathbb{R}^{|\Omega|} : w(\omega) = \mathbb{E}_{\hat{\sigma}}[u(a, \omega, \theta) | \omega] \text{ for some } \hat{\sigma} \in \hat{\Sigma}\},$$

which is a compact convex subset of finite-dimensional $\mathbb{R}^{|\Omega|}$ (boundedness of $u + \text{compact } A + \text{continuity of } u \text{ in } a$). An agent strategy is a Borel labeling $w : M \rightarrow W$, and the value of

the game depends on w only through the compact menu $C := \overline{w(M)} \subseteq W$. The space $\mathcal{K}(W)$ of nonempty compact subsets of W is itself compact under the Hausdorff metric.

This delivered a strictly stronger three-tier theorem (the v8 result, stated in the companion exposition):

- **Tier 1a** (standing alone): existence of value-optimal σ^* and ε -adversaries. **No added hypotheses.**
- **Tier 1b** (+ exact-contact): exact β^* via Borel selector on the rowwise contact set $G(s) := \{m : s \cdot w^*(m) = \min_{C^\dagger} s \cdot z\}$ where $C^\dagger := \overline{w^*(M)} \subseteq C^*$.
- **Tier 2** (+ menu-Hall): full robust rationalizability q_β -a.e. via a set-valued kernel κ supported on $G(s)$ with calibrated posterior membership in the Bayes cones $B(m)$.

The pivot eliminated A5-thick (Tier 1 became unconditional), endogenized A8c-attain (now exact-contact, checkable from the chosen labeling rather than from a representative of an abstract maximizer), and weakened TRE-gen-Hall to set-valued menu-Hall.

4 Sharpness

Two strengthenings of the v7 ternary non-radial witness were proved in v8:

Lemma 1 (Cone intersection). *For every nonempty $I \subseteq \{0, 1, 2\}$ and every Borel probability ρ on $\Delta(\Omega)$ supported on $K_I^- := \{s : s_i \leq s_k \forall i \in I, \forall k\}$ with barycenter in $B_I := \{p : p_i \geq p_k \forall i \in I, \forall k\}$: $\rho = \delta_{\mu_0}$ where $\mu_0 = (1/3, 1/3, 1/3)$.*

Theorem 2 (No-free-dust). *Under atomless τ in WTA ternary, no Borel τ -null dust set $N \subseteq M$, no Borel labeling w_N , and no adversarial kernel κ supported on rowwise minimizers can simultaneously satisfy positive $q_\beta(N)$ and Bayes-cone calibration q_N -a.e. on N .*

The cone intersection lemma upgrades sharpness from a single boundary point ($I = \{0\}$, $t_0 = (0.4, 0.3, 0.3)$) to a uniform statement across all support patterns. The no-free-dust theorem closes the escape that the q_β -a.e. reading of Definition 2 (which allows adversaries to use τ -null messages) might suggest.

Classification. Separately, the witness's trust region $T = \{\mu : \mu(0) \leq 0.4\}$ was shown not primitive: its induced payoff-profile menu is the full vertex set $\{v_0, v_1, v_2\}$, behaviorally equivalent to $T = \Delta(\Omega)$. Hence the witness exhibits a menu-engine artefact, not a counterexample to unrestricted Theorem 2.

5 Post-v8 strategic search: what we tried

After v8, three further routes were attempted to remove menu-Hall as an added hypothesis. All three reached defensible terminal endpoints without succeeding.

5.1 Strategy 3: canonical/minimal menu route

Idea: enforce a primitive minimality condition on the optimal labeled menu and ask whether menu-Hall is then derivable. Four candidate canonicity conditions were evaluated against a *renaming test*: a candidate is alive only if its primitive hypothesis does not implicitly require the existence of a kernel on $G(s)$, posterior membership in $B(m)$, or any direct Bayes-cone inclusion at messages.

- (C1) *Behavioral minimality.* C^* is F -optimal and no proper compact $D \subsetneq C^*$ is F -optimal.
- (C2) *Exposed-extreme canonicity.* C^* is the closed convex hull of its primitive aligned-best labeling.
- (C3) *Primitive trust-region minimality.* T^* is minimal among Theorem-1-equivalent trust regions; $C^* = \bigcup_{\mu \in T^*} R(\mu)$ where $R(\mu)$ is the primitive Bayes-profile correspondence.
- (C4) *Radial / orbit symmetry.* A compact group acts preserving τ , u , W ; the quotient is one-dimensional with a scalar primitive maximizer.

The critical lemma was *Lemma A.2 (uncalibrated minimal menu pruning)*: under (C1) and (exact-contact), if menu-Hall fails, then there exists a proper compact $D \subsetneq C^*$ with $F(D) \geq F(C^*)$, contradicting (C1). The contrapositive form would say: under (C1) + (exact-contact), menu-Hall holds.

Outcome. Lemma A.2 stalled at three named structural gaps under standing + (exact-contact):

1. *Label-saturation gap.* Exact-contact gives nonempty $G(s)$ but not that touching one C^* -minimizer for source s removes all of them.
2. *Replacement-index mismatch.* Hall's separating certificate uses $z(m)$ indexed by message; deletion improvement $\ell_D(s) - \ell_{C^*}(s)$ is indexed by source.
3. *Borel-to-compact gap.* $\overline{w^*(M \setminus E)} = C^*$ can hold for τ -positive Borel E (fat-Cantor type example).

5.2 Route 1: deletion-compatible Hall duality (exact theorem)

After Strategy 3 named the structural gaps, a fresh search identified the underlying missing object: a duality theorem that prices sourcewise deletion while enforcing messagewise calibration. The target theorem was formalized in primitive terms:

Hypothesis (H_{del}). For every Borel $E \subseteq M$ with $\tau(E) > 0$, if $D_E := \overline{w^*(M \setminus E)}$ is a proper compact subset of C^* , then $F(D_E) < F(C^*)$.

Conclusion. There exists a Borel kernel $\kappa : M \rightarrow \Delta(M)$ supported on $G(s)$ -a.e. with the disintegration posterior in $B(m)$ q-a.e.

A literature search across Strassen/Kellerer, weak optimal transport (Gozlan-Roberto-Samson-Tetali; Backhoff-Veraguas-Beiglböck-Pammer), martingale OT, capacity-constrained OT, adapted/causal OT, cyclic monotonicity, and persuasion duality (Doval-Skreta; Dworczak-Martini; Dworczak-Kolotilin) returned a *Build* verdict: no off-the-shelf theorem applies under primitive-equivalent hypotheses. The closest tools provide ingredients but not the finished bridge.

The natural attack was a finite-partition LP with τ -aligned mass charged sourcewise, finite Farkas duality, and a collapse lemma converting infeasibility to a positive-mass deletion certificate. The LP was attempted twice (representative-label form + sourcewise rewrite).

Outcome. The first attempt was disproved by a concrete binary counterexample: representative-label LPs generate phantom infeasibilities from coarse mixed cells unrelated to any sourcewise deletion. The sourcewise rewrite cleared that bug but stalled at four structural walls:

1. Continuum-mass label fibers can be τ -null, leaving the LP's aligned attribution empty.
2. The Borel-to-compact gap survives.
3. Finite $V_0 \subset W$ doesn't control compact max-min over W without uniform Lipschitz mesh.
4. (H_{del}) is pointwise strict, not uniformly strict. Proper $D_n \rightarrow C^*$ in Hausdorff with $F(D_n) \uparrow F(C^*)$ are consistent with (H_{del}) , so any LP conclusion of the form $F(D_E) \geq F(C^*) - K\varepsilon$ does not contradict it.

The fourth obstruction is decisive: (H_{del}) is too weak as a primitive hypothesis to generate an LP contradiction.

5.3 Route 2: calibration-defect (quantitative bound)

The fourth obstruction in Route 1 motivated a different deliverable: instead of an exact existence theorem, prove a quantitative bound

$$BR(C^*, w^*) \leq \Phi(\Delta_{\text{del}}(C^*, w^*))$$

where BR is a primitive payoff-profile Bayes-regret functional and Δ_{del} is a primitive deletion-defect functional. Under this framing, a positive defect produces positive regret rather than requiring contradiction — the (H_{del}) tiny-hinge obstruction is dodged by quantitative softness.

After three candidate defect functionals were evaluated (the chosen one, $\Delta_{\text{del}}^{\text{cp}}$, is the compact-patch dual deletion residual — primitive, non-tautological), the proof route reduced to a finite-net entropy-bootstrap with linear Φ for polytope W and Hölder $\Phi(\delta) \asymp \delta^{1/(d+1)}$ for curved W .

Outcome. The critical finite-family compact-patch separation lemma stalled at three obstructions: (i) Borel-to-compact non-monotonicity — the signed integrand $s \cdot (v_i - w^*(m))$ makes the inner $\inf_{m \in G(s)}$ non-monotone in E , so ordinary inner regularity does not bridge E_i to compact K_i ; (ii) cell-flow lift gap — LP solutions averaging over source cells cannot in general lift to Borel kernels supported on R_0 without fiber-thickness or splitability hypotheses outside the standing set; (iii) slack discipline failure — the entropy bootstrap requires $n(\varepsilon) \rho_\varepsilon \rightarrow 0$ uniformly across the dual class generated by the finite net, which standing hypotheses don't supply for curved W .

The diagnosis: Route 2 found a different scorch pattern. Route 1 died on the pointwise-strict-vs-uniform-strict gap; Route 2 dodges that wound but the remaining failures are recognizably from the same family — Borel certificates do not automatically compactify, and sourcewise deletion does not automatically lift to messagewise calibrated fibers.

6 Convergent diagnosis

Strategy 3 (Lemma A.2), Route 1 (LP collapse + sourcewise rewrite), and Route 2 (finite-net forward bound) reach the same locked gate. The single missing object is what we will call a *deletion-compatible Hall duality theorem*:

Take an optimal labeled menu (C^*, w^*) , a rowwise-minimizer correspondence $G(s) := \{m : s \cdot w^*(m) = \min_{z \in C^*} s \cdot z\}$, and Bayes-optimality cones $B(m) := \{\mu : \hat{\sigma}^*(m) \in \arg \max U(\hat{\sigma}', \mu)\}$. Give necessary and sufficient conditions *in primitive terms* (using only F , w^* , τ , α , compact source patches, raw payoff comparisons $s \cdot v$ for $v \in W$) under which there exists a Borel kernel $\kappa(\cdot \mid s)$ supported on $G(s)$ -a.e. such that the induced disintegration posterior lies in $B(m)$ for q-a.e. m .

The theorem must price sourcewise deletion (the dual character) and enforce messagewise calibration (the primal character) simultaneously, and it must explicitly handle:

- **Borel-to-compact non-monotonicity.** Signed deletion integrands prevent compact-patch tests from controlling Borel violations.
- **Label-fiber lift.** Cell-flow LP solutions to Borel kernels on the contact relation R_0 .
- **Slack discipline.** $n(\varepsilon) \rho_\varepsilon \rightarrow 0$ uniform across the dual class generated by an ε -net $V_\varepsilon \subset W$.

If such a theorem exists, Strategy 3, Route 1, and Route 2 all reopen with real force; Tier 2 may become unconditional or replace menu-Hall by a primitive defect.

7 What is banked, what is open

Banked.

- The three-tier theorem (v8 in the companion exposition): unconditional Tier 1a, exact-contact Tier 1b, set-valued menu-Hall Tier 2 in the q_β -a.e. reading of Definition 2.
- The sharpness package: cone intersection lemma + no-free-dust theorem.
- The classification of the ternary witness as a menu-engine artefact, not a counterexample to unrestricted Theorem 2.

Closed routes (not to be revisited under standing alone).

- The Phil Reny two-stage restricted-game-plus-Lusin-lift route: superseded by the menu engine, which is strictly stronger.
- Strategy 3 canonical/minimal menu (under (H_{del}) or $(C1)$ alone): blocked at three structural gaps.
- Route 1 deletion-compatible Hall as an exact theorem under (H_{del}) : blocked by pointwise-strict-vs-uniform-strict.
- Route 2 calibration-defect quantitative bound under (H_{del}) with compact-patch defect: blocked at Borel-to-compact non-monotonicity, label-fiber lift, slack discipline.

Separate research target. A Borel-residual variant of Route 2 (replace the compact-patch defect with a Borel-patch defect): the natural separation lives in the Borel world, but the cell-flow lift gap likely still bites and Φ degrades. Worth its own pipeline if pursued, not part of this project’s closure.

Open. The deletion-compatible Hall duality theorem of §6. This is the single object whose existence (or impossibility under some primitive obstruction) would resolve the unrestricted infinite extension of Theorem 2 in either direction.

References

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